

How does the Turtle Cross the Sand?



Today we're going to use Desmos to investigate the journeys of turtles crossing the sand. Go to student.desmos.com and type in the code given by your teacher. Then complete the questions below as you're going through the activity.

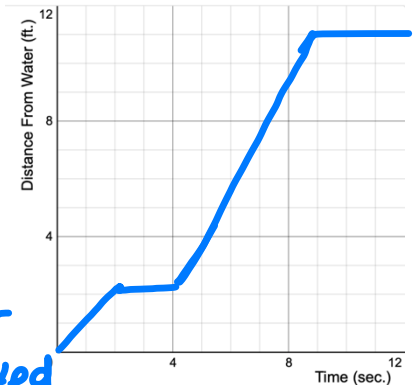
1. On **slide 1**, draw a distance vs. time graph to represent a turtle's journey crossing the sand. Then press play on the animation to see what happens. What do you notice? What do you wonder?

I notice that when I draw a horizontal line the turtle doesn't move. I notice that the steeper the graph, the faster he goes. I wonder what the equation is for my graph.

2. Copy Luca's graph on the coordinate grid shown. Then finish the story to describe this turtle's journey based on the graph.

For the first two seconds, the turtle moved away from the water at a constant rate. Then...

he didn't move for 2 seconds. Then the turtle moved away from the water at a constant rate again for just over 4 seconds. Then he stayed still.



3. a. How far away from the water was the turtle after 8 seconds?

10 feet

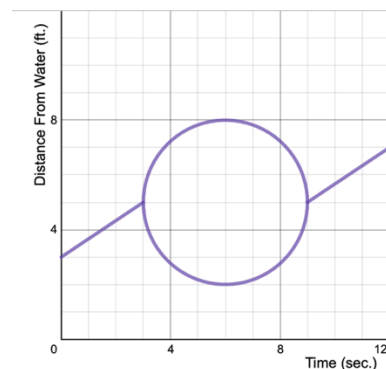
- b. At what time was the turtle 4 feet away from the water? How do you know?

After 5 seconds because the y-value when the time is 5 seconds is 4 feet.

- c. At what time was the turtle 12 feet away from the water? How do you know?

Never, the graph never reaches 12 ft.

4. Arnav's distance vs. time graph is shown below. Does this graph represent a possible journey for the turtle? Why or why not?



5. Complete **slides 7-10** on Desmos.

Is it possible for the turtle to be 5 feet away from the water at multiple points in his journey? Explain.

yes, he could have moved backwards & forwards and pass each location multiple times on the journey.

distance is a function of time $d(t)$

$d(8) = 10$

$d(t) = 4$ when $t = 5$.
Range of $d(t)$ is $[0, 11]$

Each input can have only one output.

Repeated outputs are allowed!

Lesson 1.1 – Functions and Function Notation

QuickNotes

A function is a relationship between two variables where one variable determines the other.

$$f(x) = y$$

name of function f independent variable (input) x dependent variable (output) y

Each input can have only one output!
To evaluate a function means to find the output for a given input by plugging the input into the function expression.

Check Your Understanding

1. Determine if the following relations represent a function. Give a reason for your answer.

a. $\{(0,-2), (3,5), (4,-2), (0,7)\}$

No, 0 has two outputs:
-2 and 7

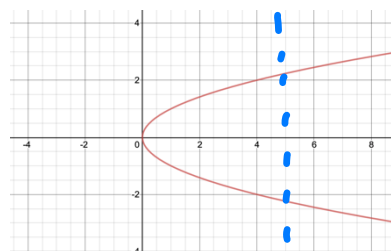
b. X is height of student; Y is his/her age

No, a height isn't associated w/ one particular age. Some 6 ft people are young, some are older.

c. X is the number of minutes on the treadmill; Y is the number of calories burned.

yes, at any moment during the workout you can have burned only one caloric amount.

d.



No, an input has multiple outputs so it fails the VLT.

2. Let $g(x) = x^2 + 5$. Find the following values:

a. $g(2) = 2^2 + 5 = 9$

b. $g(-7) = (-7)^2 + 5 = 54$

c. $g(t) = t^2 + 5$

d. $g(\text{smiley face}) = \text{smiley face}^2 + 5$

e. $g(x+h) = (x+h)^2 + 5$

3. Let $P(t)$ represent the number of people in a store, t hours after the store opens at 6 AM. Interpret each of the following statements in context.

a. $P(2) = 83$

At 8 AM there are 83 people in the store.

c. $P(L) = 121$

1 hour after the store opens, there are 121 people in the store.

b. $P(3) = P(6)$

The same amount of people are in the store at 9 AM and noon.